

MATHEMATICAL TREATISE IN NINE SECTIONS

數學九章

卷二

天時類

Heavenly Patterns: Calendar, Astronomy and Weather

秦九韶著

秦九韶 (1202–1261)

宋淳祐七年 (1247) 成書

以四庫全書本為底本，參以宜稼堂叢書本 *Expanded Edition with English Commentary*

天時類凡九問，上分五問：推氣治歷、治歷推閏、治歷演紀、綴術推星、揆日究微；
下分四問：天池測雨、圓罌測雨、峻積驗雪、竹器驗雪。

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數學九章卷二序

天時類序

夫天行有常，曆象攸繫。聖人則之，以授民時。自黃帝迎日推策，顓頊命南正重司天，堯命羲和欽若昊天，曆象日月星辰，敬授人時。三代以降，曆法屢更。漢太初、唐大衍、宋紀元，皆一時之精密者也。

曆法之要，首在測天。測天之家，必立表臬以正朝夕，候中星以定四時。冬至之日，立八尺之表，日中而測其景。景最短者，夏至也；景最長者，冬至也。二分之日，景在長短之中。此自然之數，不可易也。

月之行天，有遲有疾。朔望相交，或有薄蝕。日蝕者，月掩其光也；月蝕者，地影蔽之也。先儒謂日蝕當朔，月蝕當望，此其常也。然亦有當朔不蝕、當望不蝕者，蓋月行有黃道內外之差，地有遠近之殊，不可一概論也。

五星之行，各有遲疾順逆留伏。歲星一歲而移一宿，熒惑二歲而周一匝，鎮星二十八歲而復初，太白、辰星出入乎日，是皆有常度可循也。然古之曆家，或測之未精，或算之未密，故其推步往往有差。

今作天時之章，凡為九問。上卷論推步之術：推氣治歷以定歲餘，治歷推閏以正朔閏，治歷演紀以大衍求一術推曆元，綴術推星以步五星，揆日究微以測日景。下卷論測候之法：天池測雨、圓罌測雨以量降水，峻積驗雪、竹器驗雪以占豐歉。蓋天時之變化無窮，而數學之推測有準。學者能明於此，則可以知天道矣。

The Tian Shi Lei (天時類, “Heavenly Patterns”) is the second of nine chapters in Qin Jiushao’s *Shuxue Jiuzhang* (數學九章, Mathematical Treatise in Nine Sections). This fascicle contains **nine problems** divided into two parts, dealing with:

- Calendar computation and astronomical predictions (Problems 1–5)
- Measurement of rainfall and snowfall (Problems 6–9)

九韶原序節錄

大则可以通神明、順性命，小则可以經世務、類萬物，詎容以淺近窺哉？

1 卷二上 (Volume 2, Part 1)

1.1 第一問：推氣治歷

推氣治歷 (Tui Qi Zhi Li)

問

太史觀測天道：慶元四年戊午歲（1198），冬至三十九日九十二刻四十五分；紹定三年庚寅歲（1230），冬至三十二日九十四刻十二分。欲知嘉泰甲子歲（1204）氣骨、歲餘、斗分各幾何？

問：The Imperial Astronomer observed the heavenly way:

- In the 4th year of Qingyuan (慶元四年, 1198, year Wu-wu 戊午), the winter solstice was at 39 days, 92 *ke* (刻), 45 *fen* (分).
- In the 3rd year of Shaoding (紹定三年, 1230, year Geng-yin 庚寅), the winter solstice was at 32 days, 94 *ke*, 12 *fen*.

It is desired to find, for the intermediate year Jiatai Jiazi (嘉泰甲子, 1204): the *qi bone* (氣骨), the *year remainder* (歲餘), and the *dou fraction* (斗分).

按

按：紹定三年庚寅冬至，實為紹定四年辛卯歲之首。辛卯距戊午凡三十四年，即積年三十三也。以兩測相距之年為法，置前測之日刻分，減後測之日刻分，不減者加紀策。累加紀策，以至合天道為止。以五日上為實，實如法得歲餘。去全日，餘為斗分。以歲餘乘前測至所求中間之年數，加前測之日刻分，去滿紀策，餘為所求氣骨。

The Shaoding 3rd year Geng-yin winter solstice actually marks the beginning of Shaoding 4th year Xin-mao (辛卯). Xin-mao is 34 years from Wu-wu, giving 33 accumulated years.

答曰

氣骨：十一日三十八刻二十分八十一秒八十小分。
歲餘：五日二十四刻二十九分三十秒三十小分。
斗分：〇日二十四刻二十九分三十秒三十小分。

- Qi bone (氣骨): 11 days, 38 *ke*, 20 *fen*, 81 *miao* (秒), 80 small fractions.
- Year remainder (歲餘): 5 days, 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions.
- Dou fraction (斗分): 0 days, 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions.

術曰曰

術曰：先以兩測相距之年數為法。置前測之日刻分，以減後測之日刻分；不減者加紀策。累加紀策，以至合天道為止——以五日上為實。實如法得歲餘。去全日，餘為斗分。以歲餘乘前測至所求中間之年數，加前測之日刻分，去滿紀策，餘為所求氣骨。

First take the number of years between the two observations as the divisor. Place the earlier observation's days, *ke*, and *fen*, and subtract from the later observation's days, *ke*, and *fen*; the remainder is the rate. (If insufficient for subtraction, add the *ji-ce* 紀策). Accumulate by adding *ji-ce* repeatedly until the result is suitable for the heavenly way—using a number above 5 days as the dividend. Divide the dividend by the divisor to obtain the year remainder. Remove whole days; the remainder is the *dou fraction*. Multiply the year remainder by the number of years from the earlier observation to the desired intermediate year, add to the earlier observation's days, *ke*, and *fen*, and remove full *ji-ce*; the remainder is the desired *qi bone*.

按

按：氣骨者，冬至後夜半甲子日之日至也。歲餘者，歲周減去六甲之餘也。斗分者，歲周減去三百六十五日之餘也。此法未知歲周之確數，故以兩測氣骨之差為實，積年為法，而求得歲餘。其法雖簡，理甚精微。蓋天道运行，歲有常數，而測候之術不能無差。兩測相距之年愈多，則積差愈著，而歲餘之求亦愈密。此所以古之曆家，必累數十年之測候，而後能定曆元也。

(Editor's note:) The qi bone is the days and fractions from the winter solstice to the Jiazi day after midnight. The year remainder is the residue when the solar year length is reduced by six Jiazi cycles. The dou fraction is the residue when the solar year is reduced by 365 days. This method does not yet know the precise solar year length, hence it uses the difference between two observed qi bones as the dividend and accumulated years as divisor.

1.2 第二問：治歷推閏

治歷推閏 (Zhi Li Tui Run)

問

開禧曆，以嘉泰四年甲子歲（1204）：天正冬至一十一日乙亥四十四刻六十一分五十四秒；天正經朔一日乙丑七十五刻五十五分六十二秒。問：閏骨、閏率各幾何？

For the Kaixi calendar (開禧曆), taking the 4th year of Jiatai (嘉泰四年, 1204, year Jiazi 甲子):

- The celestial winter solstice is at 11 days, day-token Yi-hai (乙亥), 44 *ke*, 61 *fen*, 54 *miao*.
- The 11th-month new moon (經朔) is at 1 day, day-token Yi-chou (乙丑), 75 *ke*, 55 *fen*, 62 *miao*.

Question: What are the *run bone* (閏骨) and *run rate* (閏率)?

答曰

閏骨：九日六十九刻五分九十一秒，餘一百六十九分之一百二十一。
閏骨率：一十六萬三千七百七十一。

- Run bone (閏骨): 9 days, 69 *ke*, 5 *fen*, 91 *miao*, with remainder 121/(169 fractions).
- Run bone rate (閏骨率): 163,771.

術曰曰

術曰：以日法通氣朔日刻分秒，各為氣骨分、朔骨分。氣骨分以約率約之，適盡者可用；或不盡而收之在一刻以下者，亦可用。然後以減朔骨分，餘為閏骨率。以日法除之，得閏骨策。

Take the day-factor (日法) to convert the qi and new-moon days, *ke*, *fen*, and *miao* into qi-bone and new-moon-bone fractions respectively. The qi-bone fraction, when reduced by the approximation rate (約率), if exactly divisible, is usable; or if the remainder after rounding is below one *ke*, it is also usable. Then subtract from the new-moon-bone fraction; the remainder is the run bone rate. Divide by the day-factor to obtain the run bone *ce* (策).

按

按：若以朔骨分直減氣骨分，餘為九日六十九刻五分九十二秒，即閏骨策。原草僅差四十八分之一百六十九分。其所以不直減而以通分、約率、累乘累除者，為後續計算之便也。蓋曆法之設，務求精密，而精密之術，必通眾分而後可齊也。

(Editor's note:) If one directly subtracts the new moon fraction from the winter solstice fraction, the remainder is 9 days, 69 *ke*, 5 *fen*, 92 *miao*, giving the run bone *ce*. The original working only differs by 48/(169 fractions). The reason for not directly subtracting but instead using common denominators, approximation, and repeated multiplication/division is to facilitate subsequent calculations.

1.3 第三問：治歷演紀

治歷演紀 (Zhi Li Yan Ji)

問

開禧曆積年七百八十四萬一千一百八十三。欲知造術之原：調日法、求朔餘、朔率、斗分、歲率、歲閏、入元歲、入閏、朔定骨、閏泛骨、閏縮、紀率、氣元率、元閏、元數、氣等率、因率、部率、朔等數、因數、部數、朔積年，凡二十三項，各幾何？

The Kaixi calendar has an accumulated year count of 7,848,183. It is desired to know the derivation of this computation: adjusting the day-factor, finding the new-moon remainder, new-moon rate, dou fraction, year rate, year intercalation, entry-year, entry-intercalation, new-moon fixed bone, intercalation floating bone, intercalation shrinkage, ji rate, qi yuan rate, yuan intercalation, yuan number, qi equal-rate, cause-rate, *bu* rate, new-moon equal-number, cause-number, *bu* number, and new-moon accumulated year—23 items in total. What is each?

答曰

答曰：日法一萬六千九百；朔餘八千九百六十七；朔率四十九萬九千六十七；斗分二十四刻二十九分三十秒三十小分；歲率六百一十七萬二千六百八；歲閏二萬七百二十八；入元歲九千二百四十；入閏三萬一千六十八；朔定骨一十二萬八千四百四十七；閏泛骨二萬五千三百九十二；閏縮三千七百六十二；紀率六十；氣元率九千一百二十；元閏三十萬六千五百六十二；元數一十二萬；氣等率六十；因率一百五十二；部率一十一；朔等數一十七；因數五千三百九；部數八萬四千五百；朔積年七百八十四萬一千一百八十三。

(Selected key results:)

- Day-factor (日法): 16,900
- New-moon remainder (朔餘): 8,967
- New-moon rate (朔率): 499,067
- Dou fraction (斗分): equivalent to 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions
- Year rate (歲率): 6,172,608
- Year intercalation (歲閏): 20,728
- Entry-year (入元歲): 9,240
- Entry-intercalation (入閏): 31,068

術曰曰

術曰：用大衍術求之。置氣骨分、朔骨分、紀分，以為大衍之數，求其衍母、衍子、正用。以衍母除氣骨分，得積年；除朔骨分，得積月；除紀分，得積紀。以六十乘積紀，得積日。以氣骨乘紀正用，得紀總；以閏骨乘朔正用，得朔總。併二總，去滿衍母，餘為所求率實。以氣骨分除之，得曆過年數；以減積年，得未至年數。

Use the method of Dayan (大衍術) to find the qi fraction, new-moon fraction, ji fraction, and derived mother (衍母), and all the proper use-numbers (正用). Divide the derived mother by the qi fraction to obtain the accumulated years; by the new-moon fraction to obtain the accumulated months; by the ji fraction to obtain the accumulated ji. Multiply by 60 to obtain accumulated days. Multiply the qi bone by the ji proper-use to obtain the ji total; multiply the run bone by the new-moon proper-use to obtain the new-moon total. Combine the two totals, remove full derived mothers, and the remainder is the sought rate-dividend. Divide by the qi fraction to obtain the calendar-passed (歷過) years; subtract from the accumulated years to obtain the years not yet reached.

按：此問大衍求一術之全用也。大衍術者，置各數求總等，以總等約一數，不約他數，以求定母。以定母相乘，得衍母。以各定母除衍母，得衍數。以衍數滿定母去之，不滿者為奇。以奇與定母，用大衍求一術入之，求得乘率。以乘率乘衍數，得用數。各用數滿衍母去之，餘為正用。此大衍術之綱要也。治曆者，以氣朔之交會為難求，必通其分，齊其數，而後可以定元。開禧曆之積年七百八十四萬餘年，自上古甲子歲天正冬至十一月甲子朔夜半冬至起算，歷經漢、唐以迄於宋，而後得此積年之數。其術雖繁，其理甚明。學者能熟習大衍之術，則治曆推步，無往而不可矣。

This problem demonstrates the full power of Qin Jiushao's **Dayan method** (大衍求一術) for solving systems of congruences, applied here to calendar computation. The 23 quantities computed form a complete system for deriving the epoch parameters of the Kaixi calendar from astronomical observations.

1.4 第四問：綴術推星

綴術推星 (Zhui Shu Tui Xing)

問

歲星合伏段：一十六日九十分，行三度九十分，去日一十三度乃見。然後順行一百一十三日，一十七度八十三分，乃留。欲知合伏段、晨疾初段、常度、初行率、末行率、平行率各幾何？

The year-star (Jupiter, 歲星) in conjunction-hidden phase (合伏段): after 16 days 90 *fen*, it moves 3 degrees 90 *fen*, removing 13 degrees from the sun before becoming visible. Then it proceeds directly (順行) for 113 days, 17 degrees 83 *fen*, before halting (乃留). It is desired to know:

- The conjunction-hidden segment (合伏段)
- The morning-rapid initial segment (晨疾初段)
- The standard degree (常度)
- The initial motion-rate (初行率)
- The final motion-rate (末行率)
- The mean motion-rate (平行率)

What is each?

術曰

術曰：綴術推之。
合伏段者，常度為總日數，初行率為總行度。末行率等于初行率減日差乘（日數減一）。
晨疾初段者，常度為所給日數。初行率為前段之末行率。末行率等于初行率減日差乘（日數減一）。
平行率者，并初行率、末行率而半之，即為平行率。

(Method of *Zhui Shu*):

1. For the conjunction-hidden segment: The constant degree is the total number of days. The initial motion rate is the total degrees moved. The final motion rate equals the initial rate minus the daily difference (日差) multiplied by (days - 1).
2. For the morning-rapid initial segment: The constant degree is the given number of days. The initial rate is the final rate of the previous segment. The final rate equals the initial rate minus the daily difference multiplied by (days - 1).
3. For the mean motion rate: Add the initial and final rates and halve; this gives the parallel (mean) rate.

按

按：五星之行，有遲疾順逆留伏之異。此術所取，惟合伏、見段之中數，而以累加累減求逐日之度。古法之疏，於五星為尤甚。原其文義，多所隱晦。今為詳釋之如下：
歲星之行，一歲一宿，十二歲一周天。其行也，初與日合，伏而不見，是為合伏。去日一十三度，晨見東方，是為見段。順行遲疾，有晨疾、晨遲、夕遲、夕疾之段。留者，不行也。逆者，西退也。伏者，與日相近而不見也。此五星運行之大概也。

(Editor's note:) The five planets' motions involve non-uniform acceleration and deceleration. The method here takes only the middle values of the conjunction-hidden and direct-visibility segments, then uses successive addition/subtraction to derive the daily motion. The ancient method's approximation is particularly evident for the five planets. The original text's language was mostly hidden and obscure; the following detailed explanation reveals the comparison between ancient and modern precision.

Key calculations:

- Conjunction-hidden segment:
 - Constant degree: 16 days 90 *fen*
 - Initial rate: 22 *fen* 7 *miao*
 - Final rate: 21 *fen* 96 *miao*
 - Mean rate: approximately 22 *fen* (after rounding)
- Morning-rapid initial segment:
 - Constant degree: 30 days
 - Initial rate: 21 *fen* 96 *miao*
 - Daily difference: 11 *miao* 23 small-*fen* 66 small-*miao*

1.5 第五問：揆日究微

揆日究微 (Kui Ri Jiu Wei)

問

歷家測景，唯唐大衍曆為最密。本朝崇天曆，陽城測景：冬至一丈二尺七寸一分五十秒，夏至一尺四寸七分七十秒，與大衍曆合。今開禧曆，臨安府測景：冬至一丈八寸二分二十五秒，夏至九寸一分。欲知臨安夏至後幾日，其景與陽城夏至之景等？又與大衍曆景較之，所差幾何？

Among all historical shadow-measurements, only the Tang dynasty's Dayan calendar (大衍曆) was the most precise. This dynasty's Chongtian calendar (崇天曆) gives for Yangcheng (陽城):

- Winter solstice shadow: 1 *zhang* 2 *chi* 7 *cun* 1 *fen* 50 *miao*
- Summer solstice shadow: 1 *chi* 4 *cun* 7 *fen* 70 *miao*

These agree with the Dayan calendar. Now the Kaixi calendar (開禧曆) gives for Lin'an (臨安府):

- Winter solstice shadow: 1 *zhang* 8 *cun* 2 *fen* 25 *miao*
- Summer solstice shadow: 9 *cun* 1 *fen*

It is desired to find: after how many days past the summer solstice at Lin'an will the shadow equal the Yangcheng summer solstice shadow? And comparing with the Dayan calendar's shadow, by how much do they differ?

答曰

答曰：大暑後五日午中，景長一尺四寸八分八十五秒（半）。與大衍曆景較之，差六寸一分二十九秒（弱）。

Five days after the Great Heat (大暑後五日) at noon; the shadow length is 1 *chi* 4 *cun* 8 *fen* 85 *miao* (half). The shadow difference is 6 *cun* 1 *fen* 29 *miao* (less).

術曰曰

術曰：用乘法率，以節率入之。

- 一、置臨安所測冬至景長，減去夏至景長，餘為景差，以為實。
- 二、置象限度九十一度三十一分四十四秒，加一十一度二十五分二十七秒半；度化為寸，得一百二寸五千六百七十一分五十秒，以為法。
- 三、實如法而一，得〇點九六六四……；棄其餘數，自乘得節泛數：九千三百四十分。
- 四、以各節氣乘率乘之，加夏至景幕，開方得各節氣景長。

Use the multiplication-rate method, with the section-rate (節率) entering into it.

1. Place Lin'an's measured winter solstice shadow, subtract the summer solstice shadow; the remainder is the shadow-difference (景差), taken as dividend.
2. Place the 象限度 (91 degrees 31 *fen* 44 *miao*), add 11 degrees 25 *fen* 27.5 small-*fen*; treating degrees as *cun*, obtain 102 *cun* 5671.5 *fen* 50 *miao* as divisor.
3. Divide the dividend by the divisor, obtaining 0.9664...; discard the remainder and square to obtain the section floating-number (節泛數): 9,340 *fen*.
4. Multiply by the section multiplication-rates for each solar term, add the summer solstice shadow-squared, take the square root to obtain each solar term's shadow length.

按

按：二十四節氣之景長，當時實測已定，本不須算。此術故為繁難，以惑學者。細考之：象限度加一十一度有餘為法，景差以此法除之而後自乘，得節率。各節氣有乘率，以節率乘之，加夏至景冪，得各節氣之景冪。故節率者，人為抽出之數也。然其理則可通：日之行天，有黃道赤道之斜交，故冬夏之景不同。地有南北之異，故臨安與陽城之景不同。以數學推之，則雖千萬里之外，皆可預知其景之長短也。

(Editor's note:) Each solar term's shadow length was determined by actual measurement at the time and does not need computation. The method presented here deliberately creates obscurity to puzzle the reader. Detailed examination shows: the 象限度 plus 11+ degrees is taken as divisor; the shadow-difference divided by this and then squared gives the section-rate. Each solar term has a multiplication-rate; multiplying the section-rate by this and adding the summer solstice shadow-power gives each term's shadow-power. Thus the section-rate is an artificially extracted number.

2 卷二下 (Volume 2, Part 2)

卷二下序

上卷論推步之術，此卷論測候之法。天時之變，莫大於雨雪。雨澤之降，關乎農桑；雪積之厚，兆於豐歉。然測雨之法，非直以器承之而得其量也。器有圓方深淺之異，地有高低平陂之殊，必以數學通之，而後可以得平地之雨深。

古之測雨，有以圓罍、方池承之者。圓罍上小下大，方池上廣下狹，皆非平地之積也。必以其容積比之於平地之積，而後可以得雨之深淺。此四問之所由設也。學者詳之。

2.1 第六問：天池測雨

天池測雨 (Tian Chi Ce Yu)

問

今州郡皆有天池盆以測雨水。然人只知盆中得水即為雨數，不知器之形狀不同，則所受之雨亦異——故不可以盆測直以為平地之雨數也。

假令天池盆：口徑二尺八寸，底徑一尺二寸，深一尺八寸。接雨水深九寸。問：平地雨深幾何？

Nowadays every prefecture and district has a “heavenly pool” basin (天池盆) for measuring rainwater. But people only know that the water in the basin equals the rain received, and do not understand that different vessel shapes receive different amounts of rain—so one cannot directly take the basin measurement as the flat-ground rain amount.

Suppose the basin has:

- Mouth diameter (口徑): 2 *chi* 8 *cun* (28 *cun*)
- Base diameter (底徑): 1 *chi* 2 *cun* (12 *cun*)
- Depth (深): 1 *chi* 8 *cun* (18 *cun*)
- Rainwater depth inside (接雨水深): 9 *cun*

What is the equivalent flat-ground rain depth?

答曰

答曰：平地雨深三寸。

Flat-ground rain depth: 3 *cun*.

術曰曰

術曰：用圓錐術。

一、以口徑乘底徑，又各自乘，并三數。以半深乘之；又以十一乘、七除，得下積。

二、取半深加雨深，以減全深，得上深。

三、以底徑減口徑；以餘乘上深。以半深乘之，加口徑乘半深，得面率。

四、以面率除半深，得水面徑。

五、續以累乘，求得雨積；以口面冪（自乘三之）除之，得平地雨深。

Use the method for circular cones (圓錐術):

1. Multiply the mouth diameter by the base diameter, and also square each; combine these three values. Multiply by half the depth; then multiply by 11 and divide by 7 to obtain the lower volume (下積).
2. Take half the depth plus the rain depth, subtract from the total depth to obtain the upper depth (上深).

3. Subtract the base diameter from the mouth diameter; multiply the remainder by the upper depth. Multiply by half the depth, add the result to the mouth-diameter times half-depth, to obtain the surface rate (面率).
4. Divide the surface rate by half the depth to obtain the water surface diameter.
5. Continue with further multiplications to isolate the rain volume, then divide by the mouth-area (squared and tripled) to obtain the flat-ground rain depth.

草曰曰

草曰：通口徑二十八寸、底徑十二寸、深十八寸、雨深九寸，皆為分。以口徑乘底徑，得三百三十六；口徑自乘，得七百八十四；底徑自乘，得一百四十四。并之，得一千二百六十四。以半深九寸乘之，得一萬一千三百七十六。以十一乘之，得十二萬五千一百三十六；以七除之，得一萬七千八百七十六分，為下積。
半深九寸加雨深九寸，得一十八寸；以減全深十八寸，得零，即上深為零。此為雨滿天池之半也。以面率法推之，水面徑適為口徑與底徑之平均。續求雨積，以口面冪除之，得平地雨深三寸。

Mathematical summary: The heavenly pool basin is a frustum of a cone. The rain volume collected is:

$$V_{\text{rain}} = \frac{\pi h_{\text{rain}}}{3} (R^2 + Rr + r^2)$$

where R is the top radius, r is the base radius, and h_{rain} is the rain depth. Dividing by the mouth area πR^2 gives the equivalent flat-ground rain depth.

2.2 第七問：圓罍測雨

圓罍測雨 (Yuan Ying Ce Yu)

問

以圓罍受雨：口徑一尺五寸五分（一尺零五分），腹徑二尺四寸，底徑八寸，深一尺六寸。接雨水深一尺二寸。用密率入圓法，問平地雨深幾何？

Using a round vessel (圓罍) to catch rain:

- Mouth diameter (口徑): 1 *chi* 5 *fen* (10.5 *cun*)
- Belly diameter (腹徑): 2 *chi* 4 *cun* (24 *cun*)
- Base diameter (底徑): 8 *cun*
- Depth (深): 1 *chi* 6 *cun* (16 *cun*)
- Rainwater depth inside: 1 *chi* 2 *cun* (12 *cun*)

Using the “dense rate” (密率, $\pi \approx 22/7$) for circle computations, what is the flat-ground rain depth?

答曰

答曰：平地雨深一尺八寸，又七萬四千零八十八分之六千四百八十三寸。（編者校正值）

Flat-ground rain depth: 1 *chi* 8 *cun* plus $\frac{6483}{74088}$ *cun*. (Editor’s corrected value.)

術曰曰

術曰：以底徑乘腹徑，又各自乘，并三數。以罍深半之乘之；以十一乘、四十二除，得下積。取罍深半之加雨深，以減全深得，上深。以腹徑減口徑；以餘乘上深。以罍深半之乘之，加口徑乘半深，得面率。以面率乘腹率；又各自乘并之；以十一乘、四十二除，得上率。并二率（上下），得雨積總數為實。以口徑自乘三之為法。實如法得平地雨深。

Multiply the base diameter by the belly diameter, and also square each; combine these three values. Multiply by half the vessel depth; multiply by 11 and divide by 42 to obtain the lower volume.

Take half the vessel depth plus the rain depth, subtract from the total depth to obtain the upper depth. Subtract the belly diameter from the mouth diameter; multiply the remainder by the upper depth. Multiply by half the depth; add the mouth-diameter times half-depth to obtain the surface rate.

Multiply the surface rate by the belly rate; also square each and combine; multiply by 11 and divide by 42 to obtain the upper rate. Combine the two rates (upper and lower) to obtain the total rain volume as dividend. Divide by three times the squared mouth-diameter to obtain the flat-ground rain depth.

按

按：原題「用密率入圓法」，此語與求平地雨深無涉。若欲求罍中雨積，則此語為當。原草有誤，今正之。蓋圓罍之形，與天池盆異。天池盆為圓臺，上下皆圓；圓罍則口小腹大，中有腹徑，故其積之求法尤繁也。

(Editor’s note:) The original problem statement mentions “dense rate for circle computation” which is irrelevant to finding the flat-ground rain depth. If one were seeking the vessel’s rain volume, then this phrase would be appropriate. There were errors in the original computation; the above presents the corrected method.

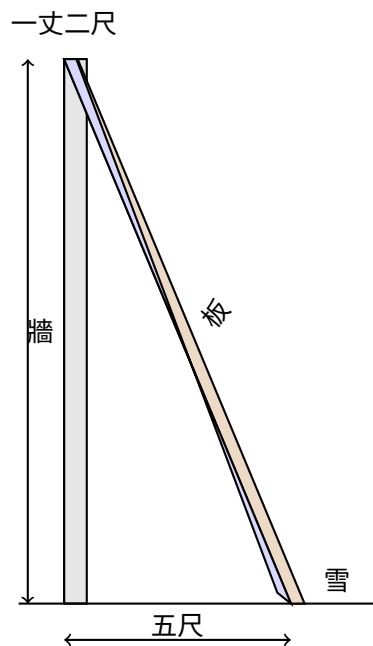
2.3 第八問：峻積驗雪

峻積驗雪 (Jun Ji Yan Xue)

問

驗雪以占豐歉：有牆高一丈二尺。有板倚之，去址五尺，板末與牆齊。雪積板上厚四寸。峻積者薄，平積者厚。問：平地雪厚幾何？

To verify snow for predicting the harvest: A wall is 1 *zhang* 2 *chi* (12 *chi*) high. A plank leans against it, the base 5 *chi* from the wall's foot, with the tip level with the wall top. Snow accumulates 4 *cun* thick on the plank. The steep accumulation is thin while the flat accumulation is thick. What is the flat-ground snow depth?



答曰

答曰：平地雪厚一尺四分。

Flat-ground snow depth: 1 *chi* 4 *fen*.

術曰曰

術曰：用少廣術，連枝入之。

一、以去址自乘，為隅。

二、以牆高自乘；并上位隅。

三、以雪厚自乘；乘以上位并數，為實。

四、(可約則約之。)開連枝平方，得平地雪厚。

Use the method of Shao Guang (少廣), with the connected-branch (連枝) method:

1. Square the distance from the base (去址): this is the corner (隅).
2. Square the wall height; combine with the corner above.
3. Square the snow thickness; multiply by the combined value above to obtain the dividend.
4. (Reduce if possible.) Extract the connected-branch square root to obtain the flat-ground snow depth.

草曰曰

草曰：通諸數為寸。去址五十寸，自乘得二千五百，為隅。牆高一百二十寸，自乘得一萬四千四百。并之：一萬四千四百加二千五百，得一萬六千九百。雪厚四寸，自乘得一十六。以上位并數一萬六千九百乘之，得二十七萬四百，為實。
隅與實俱以一百約之，得實二千七百四，隅二十五。開連枝平方： $\sqrt{2704/25} = 52/5 = 10.4$ 寸，即一尺四分。

Convert all measurements to *cun*:

- Distance from base: 50 *cun*; squared = 2,500 (corner).
- Wall height: 120 *cun*; squared = 14,400.
- Combined: 14,400 + 2,500 = 16,900.
- Snow thickness: 4 *cun*; squared = 16.
- Multiply: $16,900 \times 16 = 270,400$ (dividend).

The corner and dividend share a factor of 100; reduce to get dividend 2,704 and corner 25. Extracting the square root: $\sqrt{2704/25} = 52/5 = 10.4$ *cun* = 1 *chi* 4 *fen*.

按

按：此題理與法皆正。實即勾股術也，但不以勾股名之耳。「連枝」之號，隱其理也。試以圖明之：AB 為平地雪厚（與牆頂雪厚等），BC 為板上餘雪。三角形 ABC 與板倚牆之三角形相似。牆高為大股，去址為大勾，板為大弦。平地雪厚以勾股比例得之。其術雖隱，其理甚明。

(Editor's note:) Both the principle and method are correct. In fact, this uses the right-triangle (勾股) method, though not named as such. The "connected-branch" terminology conceals the underlying principle. The diagram illustrates: AB is the flat-ground snow depth (equal to the wall-top snow depth), BC is the excess snow on the plank. Triangle ABC is similar to the triangle formed by the plank leaning against the wall. The wall height is the large leg, the distance from base is the large base, the plank is the large hypotenuse. The flat-ground snow depth is found by proportional reasoning via the Pythagorean theorem.

2.4 第九問：竹器驗雪

竹器驗雪 (Zhu Qi Yan Xue)

問

以圓竹器驗雪：口徑一尺六寸，深一尺七寸，底徑一尺二寸。雪入其中，深一尺。竹器透風，雪多入則平地淺。問：平地雪厚幾何？

Using a round bamboo container (圓竹器) to verify snow:

- Mouth diameter (口徑): 1 *chi* 6 *cun* (16 *cun*)
- Depth (深): 1 *chi* 7 *cun* (17 *cun*)
- Base diameter (底徑): 1 *chi* 2 *cun* (12 *cun*)

Snow falls into it, filling to a depth of 1 *chi*. The container is permeable to wind; when much snow enters, the flat-ground depth is less. What is the flat-ground snow thickness?

答曰

答曰：平地雪厚九寸，又三千四百三十九分之七百六十四寸。

Flat-ground snow thickness: 9 *cun* plus $\frac{764}{3439}$ *cun*.

術曰

術曰：以底徑減口徑；以餘乘雪深，半之，自乘，為隅。以器深自乘，乘雪深自乘；并隅，乘雪深自乘，為實。開連枝三乘方，得平地雪厚。

Subtract the base diameter from the mouth diameter; multiply by the snow depth, halve, and square to obtain the corner (隅). Square the container depth and multiply by the squared snow depth; combine with the corner and multiply by the squared snow depth to obtain the dividend. Extract the connected-branch cube root (三乘方) to obtain the flat-ground snow thickness.

草曰

草曰：通諸數為寸。口徑十六寸，底徑十二寸，差四寸；雪深十寸。以差乘雪深，得四十；半之，得二十，此為上廉。器深自乘，得一十七自乘為二百八十九。雪深自乘，得一十自乘為一百。以二百八十九乘一百，得二萬八千九百；并隅續求之，開三乘方，得平地雪厚九寸，餘七百六十四分，以三千四百三十九為法，即九寸又三千四百三十九分之七百六十四寸。

Convert all measurements to *cun*:

- Mouth diameter: 16 *cun*; base diameter: 12 *cun*.
- Difference: 4 *cun*; snow depth: 10 *cun*.
- Multiply: $4 \times 10 = 40$; halve = 20 (this is the upper-*lian* 上廉).
- Container depth squared: $17^2 = 289$.
- Snow depth squared: $10^2 = 100$.
- Multiply: $289 \times 100 = 28,900$; combine with the corner and continue the cube-root extraction.

The working proceeds through the Chinese “method of extracting cube roots with carrying” (三乘方開法), yielding the flat-ground snow depth of 9 *cun* with remainder 764/3439.

按

按：「透風」之語，不關算術。或謂器中雪深與平地雪深異，以風力使然。此問之術，與天池測雨之術同旨，特以圓竹器代天池盆耳。其形亦為圓臺，而算法之用三乘方者，以器深與雪深之比例，非平方所能盡也。三乘方者，即今之所謂開立方以上之方也。秦氏之術，可謂精矣。

(Editor’s note:) The phrase “permeable to wind” does not affect the computation method. Some commentators suggest the snow depth in the container differs from the flat-ground depth due to wind effects; this problem uses the same geometric principle as the Heavenly Pool rain problem, adapted for a frustum-shaped bamboo container.

曆法天文論 (Extended Commentary)

論日蝕月蝕

日蝕者，月體掩日光也。凡日蝕必當朔，月蝕必當望，此其常也。然亦有當朔不蝕、當望不蝕者，何也？

月之行天，有黃道、白道之異。黃道者，日之軌道也；白道者，月之軌道也。二道相交，有黃白交點。月行至交點而與日會，則日蝕生焉。若月行不當交點，雖在朔日，亦不蝕也。

月蝕者，地體蔽日光，月入地影中也。月行至交點而與日對，地居日月之間，則月蝕生焉。若月行不當交點，雖在望日，亦不蝕也。故日蝕之限，在朔前後各六度有奇；月蝕之限，在望前後各十二度有奇。此界限之度數，曆家所當精測者也。

古之曆家，推步交食，有交點週期。交點退行，每十八年有餘而一周，謂之沙羅週期。一週之內，日蝕月蝕之數、之度，大略相同。此自然之數也。秦氏之大衍術，可以推此交點，可以定此週期，其用至大矣。

論五星遲疾

五星之行，各有常度，而又有遲疾順逆之變。何以言之？

歲星（木星）一歲而移一宿，約十二年一周天。其行也，在合伏之段最疾，見段之初稍遲，留則不行，逆則西退。所以然者，五星與地，同繞日而行，有內外遠近之殊，故視其行有遲疾之異。此西人第谷、開普勒之所明，而秦氏以綴術推之，亦得其近似焉。

熒惑（火星）之行，二歲而周一匝。其逆行也，每二次。鎮星（土星）之行，二十八歲而復初。太白（金星）、辰星（水星），出入乎日，或為晨星，或為夕星。太白晨見東方謂之啟明，夕見西方謂之長庚。此五星之大略也。

古之推步，以平率求之，以加減損益之。秦氏綴術之設，亦以平率為基，而以日差逐減，求其初末之率。此法雖簡，於遲疾之變，未能盡合。然在當時，已為精密矣。

論六十甲子與干支

六十甲子者，天干地支之配合也。天干十：甲乙丙丁戊己庚辛壬癸；地支十二：子丑寅卯辰巳午未申酉戌亥。以天干之首甲，配地支之首子，為甲子。以次相配，六十而周，故為六十甲子。曆法之用干支也，以紀日、紀月、紀年。日有六十之周，故日以干支紀；年有六十之周，故年以干支紀。此古今之通制也。

推氣治曆之術，以冬至為歲首，以甲子為日首。求冬至後夜半甲子之日，即求氣骨也。氣骨之數，為歲餘之積。歲餘者，歲周減六甲（三百六十日）之餘也。斗分者，歲周減三百六十五日之餘也。知歲餘與斗分，則可以推各歲之冬至，可以定曆元矣。

大衍求一術之所以為妙者，以干支之配合，有最小公倍數之關係；而以各數之互素，求其衍母、乘率，則雖千萬年之積，可以一朝而定。此秦氏之絕詣也。

天時類九問總表 (Summary of Fascicle 2 Problems)

	No.	Title	Pinyin	Topic
1	推氣治歷	Tui Qi Zhi Li		以兩測冬至之日刻分，求中間年之氣骨、歲餘、斗分
2	治歷推閏	Zhi Li Tui Run		以冬至與經朔之日刻分，求閏骨、閏率
3	治歷演紀	Zhi Li Yan Ji		以大衍求一術推開禧曆二十三項曆元之數
4	綴術推星	Zhui Shu Tui Xing		以綴術推歲星合伏、晨疾初段之常度、行率
5	揆日究微	Kui Ri Jiu Wei		以兩地測景之數，推節氣景長之差
6	天池測雨	Tian Chi Ce Yu		以圓臺形天池盆受雨，求平地雨深
7	圓罌測雨	Yuan Ying Ce Yu		以圓罌受雨，用密率求平地雨深
8	峻積驗雪	Jun Ji Yan Xue		以斜板積雪，用勾股術求平地雪厚
9	竹器驗雪	Zhu Qi Yan Xue		以圓竹器受雪，用三乘方求平地雪厚

天時類術名釋義

- 一、**氣骨**：冬至後夜半甲子日之日至也。以日刻分秒計之。
- 二、**歲餘**：歲周（一歲之日數）減六甲周期（360 日）之餘數也。
- 三、**斗分**：歲周減 365 日之餘數也。即歲餘減 5 日之餘。
- 四、**閏骨**：冬至與經朔相距之日刻分也。用以定閏月。
- 五、**閏率**：閏骨以日法通之之分數也。
- 六、**日法**：曆法之母數也。諸分皆以此為法而通之。
- 七、**朔餘**：朔策（一朔望月之日數）減 29 日之餘，以日法表示之也。
- 八、**大衍求一術**：求乘率之法，以奇定相求，得一為止。用以解一次同餘式組，為秦九韶之創造。
- 九、**綴術**：推五星遲疾之術，以逐段加減日差而求初末行率也。
- 十、**三乘方**：開立方以上之方法，即求三次方程之正根也。

End of Fascicle 2

“七政周旋於蒼穹，人事追綴以推步。
星以夜測，晷以晝量，歷年既久，則法愈疏；
惟智者能革而新之。”
— 秦九韶《數學九章序》

天時類終